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DISPLAY SYSTEM AND METHOD OF OPERATING DISPLAY SYSTEM

Abstract

The present disclosure relates to a display system and a method of operating the display system for enabling display to be continued even if any of a plurality of power supply devices for normal luminance fails in a case where a power supply device for a high-luminance direct-view light emitting diode (LED) display is implemented by the plurality of power supply devices for normal luminance. A plurality of normal-luminance power supplies that supplies power necessary for driving a driver group that drives light emitting diodes (LEDs) arranged in an array on the basis of a video signal in a normal-luminance mode, and a power supply switch that switches on/off of power supply from the plurality of normal-luminance power supplies to the driver group are provided, and when an abnormality is detected in the power supply, the power supply switch is turned off, and the LEDs are switched from a high-luminance mode to the normal luminance mode and are driven, and then the power supply switch is controlled to be turned on. The present disclosure can be applied to an LED display device.

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Background/Summary

TECHNICAL FIELD
[0001] The present disclosure relates to a display system and a method of operating the display system, and more particularly to a display system and a method of operating the display system for enabling display to be continued even if any of a plurality of power supply devices for normal luminance fails in a case where a power supply device for a high-luminance direct-view light emitting diode (LED) display is implemented by the plurality of power supply devices for normal luminance.
BACKGROUND ART
[0002] A direct-view display market using a light emitting diode (LED) as a display element is expanding, and various technologies related to such an LED display have been proposed.
[0003] For example, a technique for suppressing occurrence of color unevenness in a direct-view display using an LED has been proposed (see Patent Document 1).
CITATION LIST
Patent Document
[0004] Patent Document 1: WO 2018/164105 A
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
[0005] Moreover, in recent years, a tiling display market in which a plurality of display units including LED displays are arranged in an array is expanding among direct-view displays using LEDs.
[0006] Furthermore, under such circumstances, development of an LED display corresponding to high luminance for the purpose of more faithfully reproducing luminance is in progress.
[0007] Moreover, since the power supply devices have also been downsized in recent years, there is a case where two power supply devices are mounted on one display unit as measures against an increase in a power supply load and a failure, and both measures against an overload and a redundancy function (a function to maintain an operation even if one of the two power supply devices fails) are achieved.
[0008] Therefore, by applying such a configuration, it is conceivable to increase a power supply capacity by operating two power supply devices of a display unit corresponding to normal luminance in parallel by power supply load distribution, and to implement a power supply device of a display unit corresponding to high luminance.
[0009] However, in a case where the power supply device of a display unit corresponding to high luminance is implemented by the two power supply devices corresponding to normal luminance as described above, when one of the two power supply devices fails, there is a possibility that the display cannot be continued due to an overload due to a shortage of the entire power even in a state where the other power supply device does not fail.
[0010] The present disclosure has been made in view of such a situation, and enables display to be continued even if any of a plurality of power supply devices for normal luminance fails in a case where a power supply device for a high-luminance direct-view light emitting diode (LED) display

is implemented by the plurality of power supply devices for normal luminance.
Solutions to Problems
[0011] A display system according to one aspect of the present disclosure is a display system including: a driver configured to drive a display element on the basis of a video signal; a plurality of power supplies configured to supply power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range; and a driver control unit configured to control the driver so as to switch a luminance mode of the display element, in which the driver control unit controls the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by power supply from the plurality of power supplies, and performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of detection of an abnormality in any of the plurality of power supplies.
[0012] A method of operating a display system according to one aspect of the present disclosure is a method of operating a display system including a driver that drives a display element on the basis of a video signal, a plurality of power supplies that supplies power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range, and a driver control unit that controls the driver so as to switch a luminance mode of the display element, the method including: by the driver control unit, controlling the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by the power supply from the plurality of power supplies; and performing control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of detection of an abnormality in any of the plurality of power supplies, in which the driver control unit drives the display element in the second luminance mode by the power supply from the plurality of power sources, and performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of the detection of the abnormality in any of the plurality of power supplies.
[0013] In one aspect of the present disclosure, a display element is driven by a driver on the basis of a video signal, power capable of driving the display element is supplied by a plurality of power supplies to the driver in a first luminance mode in which the display element is driven in a first luminance range, the driver is controlled by a driver control unit so as to switch a luminance mode of the display element, the driver is controlled by the driver control unit to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by power supply from the plurality of power supplies, and control is performed by the driver control unit to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of detection of an abnormality in any of the plurality of power supplies.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0014] FIG. **1** is a diagram for describing respective power supplies of a normal-luminance display and a high-luminance display.
[0015] FIG. **2** is a diagram for describing a configuration example of a display unit that implements a power supply of a high-luminance display by two normal-luminance power supplies.
[0016] FIG. **3** is a diagram for describing an outline of a display unit of the present disclosure.

[0017] FIG. **4** is a diagram for describing a configuration example of a favorable embodiment of the display system of the present disclosure.
[0018] FIG. **5** is a diagram for describing a configuration example of a video wall controller in FIG. **4**.
[0019] FIG. **6** is a diagram for describing a configuration example of a display unit in FIG. **4**.
[0020] FIG. **7** is a flowchart for describing display processing by the display system in FIG. **4**.
[0021] FIG. **8** is a flowchart for describing driver control processing by the display system in FIG. **4**.
[0022] FIG. **9** is a diagram for describing an application example of the display system of the present disclosure.
[0023] FIG. **10** is a flowchart for describing abnormality detection signal switch control processing by the display system in FIG. **9**.

MODE FOR CARRYING OUT THE INVENTION

[0024] Hereinafter, a favorable embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. Note that, in the present specification and the drawings, components having substantially the same functional configurations are denoted by the same reference signs, and redundant descriptions are omitted.
[0025] Hereinafter, a mode for carrying out the present technology will be described. Description will be given in the following order. [0026] 1. Outline of Present Disclosure [0027] 2. Favorable Embodiment [0028] 3. Application Example

1. Outline of Present Disclosure

[0029] The present disclosure enables display to be continued even if an abnormality occurs in any of a plurality of power supply devices for normal luminance in a case where a power supply device for a high-luminance direct-view light emitting diode (LED) display is operated by the plurality of power supply devices for normal luminance.
[0030] In recent years, a tiling display market in which a plurality of display units including small LED displays is arranged in an array is expanding among direct-view displays using LEDs, and moreover, development of high luminance for the purpose of more accurate luminance reproduction has been advanced.
[0031] More specifically, there has been a high-luminance model such as several 1000 cd/m.sup.2 in a direct-view LED display for outdoor use having a dot pitch of about 10 mm.
[0032] However, in recent years, miniaturization of LED chips has progressed, and direct-view LED displays having a dot pitch of around 1.0 mm have rapidly spread, and luminance of these LED displays is approximately 1000cd/m.sup.2.
[0033] This luminance is a result of being restricted by heat dissipation, a power supply capacity (power supply size) to be mounted, and by increasing the power supply capacity, a direct-view LED display with ultra-high luminance of, for example, 4000 cd/m.sup.2 or more, which greatly exceeds 1000 cd/m.sup.2, has appeared although it depends on the use.
[0034] To implement high luminance of the LED display, a large-capacity power supply LB having a larger capacity than that of a normal-luminance power supply NB as illustrated in the upper part of the lower row of FIG. **1** is required for the normal-luminance power supply NB required for a normal-luminance display ND as illustrated in the upper row of FIG. **1**.
[0035] That is, the upper part of the lower row of FIG. **1** illustrates a configuration in which the large-capacity power supply LB having a larger capacity than that of the normal-luminance power supply NB supplies power to a high-luminance display HD including an LED display corresponding to high luminance.
[0036] However, simply increasing the capacity of the power supply increases the size of a housing constituting the display unit as the large-capacity power supply LB increases in size.
[0037] Furthermore, in a case where the power supply of the display unit is configured only with the large-capacity power supply LB, display of the high-luminance display HD cannot be continued

when a failure occurs.

[0038] Moreover, when the display units including the high-luminance display HD are connected in a row (in a daisy chain), a signal cannot be transmitted to the display unit in a subsequent stage, and thus there is a possibility that the display unit int the subsequent stage cannot be displayed.

[0039] Meanwhile, since the normal-luminance power supply has been downsized through recent development, it is conceivable to supply power to the high-luminance display HD by increasing the capacity by two normal-luminance power supplies NB-**1** and NB-**2** as illustrated in the lower part of the lower row of FIG. **1**.

[0040] Moreover, by increasing the capacity by the two normal-luminance power supplies NB-**1** and NB-**2**, for example, even if an abnormality occurs in one of the two normal-luminance power supplies NB, the other normal-luminance power supply NB can continue the display with the normal luminance.

[0041] Note that the number of the normal-luminance power supplies NB may be two or more as long as it is plural. Furthermore, the “plurality of normal-luminance power supplies NB” may include a case where one power supply device includes a plurality of power supply regions capable of individual operation corresponding to the normal-luminance power supply NB.

<Configuration Example in Case of Using Two Normal-Luminance Power Supplies>

[0042] FIG. **2** illustrates a configuration example of a display unit DU that supplies power corresponding to high luminance to a driver IC group DG that drives LEDs corresponding to high luminance by the two normal-luminance power supplies NB-**1** and NB-**2**.

[0043] The display unit DU of FIG. **2** includes normal- luminance power supplies NB-**1** and NB-**2**, diodes D-**1** and D-**2**, a driver control circuit DC, and the driver IC group DG.

[0044] Each of the normal-luminance power supplies NB-**1** and NB-**2** is a power supply device used in a display unit including an LED display corresponding to normal luminance. The normal-luminance power supplies NB-**1** and NB-**2** are provided with terminals for receiving an AC input from an alternating current (AC) power supply, and are respectively provided with power supply terminals F-**1** and F-**2** for supplying power to the driver control circuit DC and the driver IC group DG via the diodes D-**1** and D-**2**. Furthermore, the normal-luminance power supplies NB-**1** and NB-**2** are respectively provided with terminals G-**1** and G-**2** connected to a ground potential GND.

[0045] The diodes D-**1** and D-**2** are favorably diode circuits using metal-oxide-semiconductor field-effect transistor (MOS-FETs) or the like because power loss is increased in simple diodes.

[0046] The driver control circuit DC receives a video signal for causing an individual LED constituting the LED display to emit light, and drives a driver IC for causing the individual LED constituting the corresponding driver IC group DG to emit light.

[0047] In the display unit DU of FIG. **2**, in a case where any of the normal-luminance power supplies NB-**1** and NB-**2** has an abnormality and does not function, the diodes D-**1** and D-**2** respectively separate the normal-luminance power supplies from the driver control circuit DC and the driver IC group DG, so that the power supply can be continued in a range of the normal luminance by the other having no abnormality.

[0048] However, due to a simple circuit configuration, the power capacity for driving the high-luminance LED cannot be interchanged, and the power supply to the driver IC group DG (LEDs) remains in the power supply corresponding to normal luminance.

[0049] That is, in a case where an abnormality occurs in either one, power interchange corresponding to high luminance cannot be performed regardless of the occurrence of an abnormality although power supply redundancy (power supply by only one of the two normal-luminance power supplies NB in the case where the other is abnormal) can be achieved.

[0050] Moreover, in a case of adding a special balance circuit (not illustrated) to implement a large capacity corresponding to high luminance by two normal-luminance power supplies, the two normal-luminance power supplies can function as a large capacity power supply. However, since a configuration is substantially equivalent to a circuit configuration in which a resistor is inserted to balance voltage, a power loss is large. In addition, when an abnormality occurs in one of the normal-luminance power supplies and the power is supplied only by the other normal-luminance power supply, the other one normal-luminance power supply needs to cover the power necessary for the high luminance. Therefore, there is a possibility that overpower (overcurrent) occurs and the operation of the display unit stops.

[0051] Therefore, the present disclosure performs sharing control of the two normal-luminance power supplies so as to have the same current value, and supplies the power in parallel, thereby implementing the power supply corresponding to high luminance.

[0052] Furthermore, each of the two normal-luminance power supplies has a function to detect an abnormal operation when a failure occurs, and outputs an abnormality detection signal when detecting an abnormality, and when the abnormality detection signal is output from one of the two normal-luminance power supplies, the power supply is temporarily stopped by turning off a power supply switch that stops the power supply to the driver IC group DG. After a light emission mode is switched from a high-luminance mode to a normal-luminance mode, the power supply switch is turned on to resume the power supply to enable the display to be continued in the normal-luminance mode, whereby the power supply redundancy is implemented.

[0053] More specifically, as illustrated in FIG. **3**, a display unit DU′ of the present disclosure includes normal-luminance power supplies NB′-**1** and NB′-**2**, a driver control circuit DC′, a driver IC group DG′, and a power supply switch SW.

[0054] Note that the normal-luminance power supplies NB′-**1** and NB′-**2**, the driver control circuit DC′, and the driver IC group DG′ of the display unit DU′ in FIG. **3** have configurations corresponding to the normal-luminance power supplies NB-**1** and NB-**2**, the driver control circuit DC, and the driver IC group DG of the display unit DU in FIG. **2**, respectively, and basically have the same functions.

[0055] That is, the display unit DU′ of FIG. **3** is different from the display unit DU of FIG. **2** in that the diodes D-**1** and D-**2** are omitted, the power supply switch SW is added, and the normal-luminance power supplies NB′-**1** and NB′-**2** are subjected to sharing control so as to have same current value and supply the power in parallel to function as the power supplies corresponding to high luminance.

[0056] Moreover, the normal-luminance power supplies NB′-**1** and NB′-**2** are respectively provided with terminals F′-**1** and F′-**2** and terminals G′-**1** and G′-**2** corresponding to the terminals F-**1** and F-**2** for supplying power of the normal- luminance power supplies NB-**1** and NB-**2** and the terminals G-**1** and G-**2** connected to the ground in FIG. **2**, and are newly provided with terminals E′-**1** and E′-**2** for outputting the abnormality detection signal.

[0057] When detecting that any abnormality has occurred in each of the normal-luminance power supplies NB′-**1** and NB′-**2**, the normal-luminance power supply NB′-**1** or NB′-**2** supplies the abnormality detection signal to the power supply switch SW and the driver control circuit DC′ from terminal E′-**1** or E′-**2**.

[0058] When the abnormality detection signal is supplied from any of the terminals E′-**1** and E′-**2** of the normal-luminance power supplies NB′-**1** and NB′-**2**, the power supply switch SW turns off a connection state to stop the power supply from the terminals F′-**1** and F′-**2** of the normal-luminance power supplies NB′-**1** and NB′-**2** to the driver IC group DG′.

[0059] Furthermore, when receiving supply of a re-energization signal from the driver control circuit DC′ after turning off the connection by the supply of the abnormality detection signal, the power supply switch SW turns on the connection to resume the power supply from the terminals F′-**1** and F′-**2** to the driver IC group DG′.

[0060] When the abnormality detection signal is supplied from any of the terminals E′-**1** and E′-**2** of the normal- luminance power supplies NB′-**1** and NB′-**2**, the driver control circuit DC′ switches the light emission mode, which is the operation mode of light emission control of the driver IC group DG′, from the high-luminance mode of supplying the power corresponding to high

luminance to the normal-luminance mode of performing the light emission control with normal luminance, and then supplies the re-energization signal to the power supply switch SW.
[0061] The light emission mode is a mode of a luminance range when the LED emits light, and includes the normal-luminance mode and the high-luminance mode. The high-luminance mode is a mode of emitting light in the luminance range wider on a high luminance side than the luminance range when light emission is controlled in the normal-luminance mode. The present disclosure performs the sharing control so as to have the same current value output from both the normal-luminance power supplies NB′-**1** and NB′-**2** unless there is abnormality in both the normal-luminance power supplies, thereby implementing a large-capacity power supply, and implementing light emission in the high-luminance mode.
[0062] That is, with the configuration as illustrated in FIG. **3**, in the present disclosure, when the abnormality detection signal is detected from any of the two normal-luminance power supplies NB′-**1** and NB′-**2**, the power supply switch SW is turned off, and the power supply to the driver IC group DG′ is temporarily stopped to stop the display.
[0063] Moreover, after the light emission mode is switched from the high-luminance mode to the normal-luminance mode by the driver control circuit DC′ in the state where the display is stopped, the re-energization signal is supplied to the power supply switch SW, and the power supply switch SW is turned on, so that the display in the normal-luminance mode is resumed.
[0064] Here, as described above, when the abnormality detection signal is detected from the terminal E′-**1** or E′-**2** of the normal-luminance power supply NB′-**1** or NB′-**2**, the power supply switch SW is turned off, and the power supply is stopped, so that the display is temporarily stopped.
[0065] However, a period required after the light emission mode is switched from the high-luminance mode to the normal-luminance mode by the driver control circuit DC′, the re-energization signal is supplied to the power supply switch SW, and the power supply switch is turned on to resume the power supply is about several frames, and thereafter, the display in the normal-luminance mode is resumed, so that the display is substantially continued. More specifically, in a case where a frame rate is 60 fps, when the period required after the power supply is stopped to when the power supply is resumed is about 100 ms, for example, the display is stopped for about 6 frames, and the display may be seen by a user as flickering for a moment, but the display does not seem to be stopped, and the display state is continued.
[0066] Thereby, it is possible to implement the power supply in the high-luminance mode while suppressing an increase in size of the housing, and to continue the display in the normal-luminance mode even in the case where an abnormality occurs in any of the plurality of normal-luminance power supplies, and to implement the power supply redundancy.
2. Favorable Embodiment
<Configuration Example of Display System>
[0067] The present disclosure enables display to be continued even if an abnormality occurs in any of a plurality of power supply devices for normal luminance in a case where a power supply device for a high-luminance direct-view light emitting diode (LED) display is operated by the plurality of power supply devices for normal luminance.
[0068] FIG. **4** illustrates a configuration example of a display system to which the technology of the present disclosure is applied.
[0069] A display system **11** in FIG. **4** displays video content on a large display (tiling display) configured by arranging a plurality of display units in an array.
[0070] More specifically, the display system **11** includes a personal computer (PC) **30**, a video server **31**, a video wall controller **32**, and a video wall **33**.
[0071] The personal computer (PC) **30** is a general-purpose computer, and receives an operation input by the user and supplies a command corresponding to a detail of the operation to the video wall controller **32**. Furthermore, when an abnormality occurs in any of display units **51-1** to **51-*n***

constituting the video wall **33**, the PC **30** acquires information of the occurrence of an abnormality via the video wall controller **32** and presents the information to the user.
[0072] The video server **31** includes, for example, a server computer or the like, and supplies data of a video signal such as video content to the video wall controller **32**.
[0073] The video wall controller **32** operates in response to the command supplied from the PC **30**, and distributes the data of the video signal of the video content among the display units **51-1** to **51-***n* constituting the video wall **33** to cause the display units **51-1** to **51-***n* to display the data.
[0074] Note that, in a case where there is no need to individually distinguish the display units **51-1** to **51-***n*, the display units **51-1** to **51-***n* are simply referred to as display unit(s) **51**.
[0075] As illustrated in the upper right part of FIG. **4**, the video wall **33** includes the display units **51-1** to **51-***n* arranged in an array, the display units **51-1** to **51-***n* each having pixels including LEDs arranged in an array, and images individually displayed by the display units **51** are combined in a tile so that one entire image is displayed on the video wall **33**.
[0076] The video wall controller **32** applies predetermined signal processing to the data of the video signal of the video content supplied from the video server **31**, distributes and supplies the data in accordance with the arrangement of the display units **51-1** to **51-***n*, controls their respective displays of the display units **51-1** to **51-***n*, and controls the video wall **33** to display one entire image.
[0077] Note that the video wall controller **32** and the video wall **33** may be integrated with each other, or may be integrated into a display device (information processing system).
<Detailed Configuration of Video Wall Controller>
[0078] Next, a detailed configuration example of the video wall controller **32** will be described with reference to FIG. **5**.
[0079] The video wall controller **32** includes a local area network (LAN) terminal **71**, a high definition multimedia interface (HDMI) (registered trademark) terminal **72**, a display port (DP) terminal **73**, a digital visual interface (DVI) terminal **74**, a network interface (IF) **75**, a micro processor unit (MPU) **76**, a signal input IF **77**, a signal processing unit **78**, a dynamic random access memory (DRAM) **79**, a signal distribution unit **80**, and signal IFs **81-1** to **81-***n*.
[0080] The local area network (LAN) terminal **71** is, for example, a connection terminal of a LAN cable or the like, and the LAN terminal **71** establishes communication with the personal computer (PC) **30** over a LAN, the personal computer (PC) **30** being operated by the user to supply a control command or the like corresponding to a detail of the operation to the video wall controller **32**, and supplies the input control command or the like to the MPU **76** via the network IF **75**.
[0081] Note that the LAN terminal **71** may have a configuration adapted to physical connection with a wired LAN cable, or may have a configuration adapted to connection with a so-called wireless LAN implemented by wireless communication.
[0082] The MPU **76** receives the input of the control command supplied from the PC **30** via the LAN terminal **71** and the network IF **75**, and supplies a control signal corresponding to the received control command to the signal processing unit **78**.
[0083] The HDMI terminal **72**, the DP terminal **73**, and the DVI terminal **74** each serves as an input terminal of the data of the video signal, and is connected to, for example, a server computer functioning as the video server **31**, and supplies the data of the video signal to the signal processing unit **78** via the signal input IF **77**.
[0084] Note that, although FIG. **2** illustrates an example where the video server **31** and the HDMI terminal **72** are connected, any of the HDMI terminal **72**, the DP terminal **73**, or the DVI terminal **74** may be selected and connected as necessary because the HDMI terminal **72**, the DP terminal **73**, and the DVI terminal **74** have different standards but basically have similar functions. Furthermore, the terminal is not limited to the HDMI, DP, and DVI standards, and a terminal according to a standard other than the HDMI, DP, and DVI standards may be provided, and any of them may be selected and connected.

[0085] The signal processing unit **78** adjusts color temperature, contrast, brightness, and the like of the data of the video signal supplied via the signal input IF **77** on the basis of the control signal supplied from the MPU **76**, and supplies the data to the signal distribution unit **80**. At this time, the signal processing unit **78** develops the data of the video signal in the connected DRAM **79**, executes signal processing based on the control signal, and supplies a result of the signal processing to the signal distribution unit **80** as necessary.
[0086] The signal distribution unit **80** distributes the data of the video signal subjected to the signal processing and supplied from the signal processing unit **78** and individually distributes and transmits the data to the display units **51-1** to **51-***n* via the signal IFs **81-1** to **81-***n*.
[0087] When executing the signal processing based on the control signal, the signal processing unit **78** executes the signal processing, assuming that the light emission mode is the high-luminance mode by default, that is, so as to obtain the luminance range corresponding to high luminance, and supplies the signal processing result to the signal distribution unit **80**.
[0088] Furthermore, when the video wall **33** notifies that an abnormality is detected in any of the power supplies of the display unit **51** via the signal IF **81** and the signal distribution unit **80**, the signal processing unit **78** switches the light emission mode from the high-luminance mode to the normal-luminance mode, executes the signal processing, and supplies the signal processing result to the signal distribution unit **80**.
[0089] Moreover, when the video wall **33** notifies that an abnormality is detected in any of the power supplies of the display unit **51**, the signal processing unit **78** supplies the information of the notification to the PC **30** via the MPU **76**, the network IF **75**, and the LAN terminal **71**. At this time, the PC **30** presents to the user that an abnormality has occurred in any of the power supplies of the display unit **51** from the video wall **33**.
[0090] Note that, in a case where an abnormality has occurred in all the power supplies provided in any of the display units **51** of the video wall **33**, the display on the display unit **51** is disabled. Therefore, the PC **30** may present that the display is disabled, that is, a black screen is displayed. Furthermore, the presentation based on the occurrence of an abnormality in the power supply may be performed by the video wall controller **32**.
<Detailed Configuration of Display Unit>
[0091] Next, a detailed configuration example of the display unit **51** will be described with reference to FIG. **6**.
[0092] The display unit **51** includes an AC input **90**, a driver control circuit **91**, an LED block **92**, normal-luminance power supplies **93-1** and **93-2**, and a power supply switch **94**.
[0093] The driver control circuit **91** is driven by a system power supply output supplied from terminals **93***s***-1** and **93***s***-2** of the power output from the normal-luminance power supplies **93-1** and **93-2**. The driver control circuit **91** supplies the data of the video signal for controlling light emission of LEDs constituting LED arrays **122-1** to **122-**N to a plurality of LED drivers **121-1** to **121-**N constituting the LED block **92**.
[0094] In a case where the abnormality detection signal is not output from any of the normal-luminance power supplies **93-1** and **93-2**, the driver control circuit **91** sets the light emission mode to the high-luminance mode, and supplies, to the plurality of LED drivers **121-1** to **121-**N constituting the LED block **92**, the data of the video signal for emitting light of the LEDs constituting the LED arrays **122-1** to **122-**N with high luminance in a wider luminance range on a high luminance side than the normal luminance.
[0095] In a case where the abnormality detection signal is output from at least one of the normal-luminance power supplies **93-1** and **93-2**, the driver control circuit **91** switches the light emission mode from the high-luminance mode to the normal-luminance mode of emitting light with normal luminance, and supplies, to the plurality of LED drivers **121-1** to **121-**N constituting the LED block **92**, the data of the video signal for emitting light of the LEDS constituting the LED arrays **122-1** to **122-**N with normal luminance.

[0096] Furthermore, in the case where the abnormality detection signal is output from at least one of the normal-luminance power supplies **93-1** and **93-2**, the power supply switch **94** controls the connection to be turned off to stop the driving of the LED block **92**. At this time, after changing the light emission mode from the high-luminance mode to the normal-luminance mode, the driver control circuit **91** controls the power supply switch **94** to be turned on to resume the driving of the LED block **92**.
[0097] More specifically, the driver control circuit **91** includes a monitoring unit **110**, a signal IF **111**, a signal processing unit **112**, a DRAM **113**, and output IFs **114-1** to **114**-N.
[0098] The signal IF **111** receives the input of the data of the video signal supplied from the video wall controller **32** and supplies the data to the signal processing unit **112**.
[0099] The signal processing unit **112** corrects color and luminance for each display unit **51** on the basis of the data of the video signal supplied from the signal IF **111**, generates data for setting light emission intensity of each LED constituting the LED arrays **122-1** to **122**-N, and distributes and supplies the data to the LED drivers **121-1** to **121**-N of the LED block **92** via the output IFs **114-1** to **114**-N.
[0100] The LED block **92** includes the LED drivers **121-1** to **121**-N and the LED arrays **122-1** to **122**-N.
[0101] The LED drivers **121-1** to **121**-N perform control of the light emission of the LEDs arranged in an array constituting the corresponding LED arrays **122-1** to **122**-N on the basis of the data for setting the light emission intensity of the LEDs **141**, the data containing the video signal supplied from the driver control circuit **91**.
[0102] The AC input **90** supplies power of an alternating current (AC) power supply to each of the normal-luminance power supplies **93-1** and **93-2**.
[0103] Each of the normal-luminance power supplies **93-1** and **93-2** is a power supply device used in the display unit including the LED display corresponding to the normal luminance, and the sharing control is performed so as to have the same current value to be output on the basis of the AC input **90**. The power is supplied from each of the terminals **93***s***-1** and **93***s***-2** to the driver control circuit **91**, and the power is supplied from each of terminals **93***d***-1** and **93***d***-2** to the LED block **92** via the power supply switch **94**.
[0104] When detecting its own abnormality, the normal-luminance power supply **93-1** or **93-2** outputs the abnormality detection signal from a terminal **93***e***-1** or **93***e***-2** to the driver control circuit **91** and the power supply switch **94**.
[0105] More specifically, the normal-luminance power supplies **93-1** and **93-2** respectively include abnormality detection units **131-1** and **131-2**, driver power supply output units **132-1** and **132-2**, and system power supply output units **133-1** and **133-2**.
[0106] Each of the abnormality detection unit **131-1** and **131-2** monitors the power supply of the normal-luminance power supply **93-1** or **93-2**, and temperature and occurrence of abnormality such as ignition, and outputs the abnormality detection signal as a status signal to the driver control circuit **91** and the power supply switch **94** from the terminal **93***e***-1** or **93***e***-2** when abnormality is detected.
[0107] Each of the driver power supply output units **132-1** and **132-2** supplies a driver power supply output from the terminal **93***d***-1** or **93***d***-2** to the driver control circuit **91** via the power supply switch **94**.
[0108] Therefore, the power is supplied to the driver control circuit **91** when the power supply switch **94** is turned on, and the power supply to the driver control circuit **91** is stopped when the power supply switch **94** is turned off, by the driver power supply outputs from the terminals **93***d***-1** and **93***d***-2**.
[0109] Each of the system power supply output units **133-1** and **133-2** supplies a system power supply output from the terminal **93***s***-1** or **93***s***-2** to the driver control circuit **91**.
[0110] Note that each of the normal-luminance power supplies **93-1** and **93-2** is a power supply

device used in the display unit including the LED display corresponding to the normal luminance. Therefore, in a case where the normal-luminance power supply is used alone, sufficient power supply can be performed when the display unit **51** of the present disclosure is used in the normal-luminance mode, but the power supply capacity is insufficient depending on a luminance level of the video signal and there is a possibility that the display unit **51** goes down due to overcurrent when the display unit **51** is used in the high-luminance mode.
[0111] The power supply switch **94** is a switch that controls on/off, such as a MOS-FET, for example, and provided on the path that connects the terminals **93*d*-1** and **93*d*-2** of the normal-luminance power supplies **93-1** and **93-2** and the driver control circuit **91** and to which the driver power supply output is supplied.
[0112] The power supply switch **94** controls the connection to be on as long as the normal state in which the abnormality detection signal is not supplied from the terminal **93*e*-1** or **93*e*-2** is continued at the same time as the power supply is turned on, and controls the connection to be off when the abnormality detection signal is supplied from the terminal **93*e*-1** or **93*e*-2** and shifts to the abnormal state.
[0113] The power supply switch **94** controls itself to be turned off, continues the off state after shifting to the abnormal state, and thereafter controls itself to be turned on again when a re-energization signal is supplied from the driver control circuit **91** and returns to the normal state.
[0114] The monitoring unit **110** monitors whether or not the abnormality detection signal is supplied from the terminals **93*e*-1** and **93*e*-2**.
[0115] When the abnormality detection signal is supplied from the terminal **93*e*-1** or **93*e*-2**, the monitoring unit **110** detects the abnormal state and notifies the signal processing unit **112** that an abnormality has occurred in the normal-luminance power supply **93-1** or **93-2**.
[0116] When the occurrence of abnormality in the normal-luminance power supply **93-1** or **93-2** is notified, the signal processing unit **112** switches the light emission mode from the high-luminance mode to the normal-luminance mode in which light is emitted with normal luminance. That is, the signal processing unit **112** switches the light emission mode from the high-luminance mode to the normal-luminance mode and generates the data in generating the data for setting the light emission intensity (luminance) of each LED constituting the LED arrays **122-1** to **122-**N, and distributes and supplies the data to the LED drivers **121-1** to **121-**N of the LED block **92** via the output IFs **114-1** to **114-**N.
[0117] That is, when the abnormality detection signal is supplied from the terminal **93*e*-1** or **93*e*-2** to be in the abnormal state, the power is not supplied from at least one of the normal-luminance power supplies **93-1** and **93-2**. Therefore, when the light of the LED is to be emitted in the luminance range in the high-luminance mode, there is a possibility that the light cannot be emitted due to overcurrent. Therefore, by switching the light emission mode from the high-luminance mode to the normal-luminance mode, the light emission intensity (luminance) is set to the luminance range of the normal-luminance mode, and driving can be performed with the power supplied from one normal-luminance power supply **93**.
[0118] The monitoring unit **110** supplies the re-energization signal to the power supply switch **94** at timing after the light emission mode in the signal processing unit **112** is switched from the high-luminance mode to the normal-luminance mode, for example, at timing after a period of about several frames.
[0119] The power supply switch **94** turns on the operation on the basis of the re-energization signal from the monitoring unit **110** of the driver control circuit **91** to resume the power supply to the LED block **92** and resume the light emission of the LEDs in the LED block **92**.
[0120] At this time, the light emission of the LEDs in the LED block **92** continues in the normal-luminance mode.
[0121] With such a configuration, even in the case where an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, the power supply to the LED block **92** is stopped

by turning off the power supply switch **94**. Therefore, even if the driver control circuit **91** controls the light emission in the high-luminance mode, it is possible to avoid a situation where the display unit goes down due to overpower (overcurrent).
[0122] Furthermore, after the light emission mode is switched from the high-luminance mode to the normal-luminance mode in the state where the power supply switch **94** is turned off and the power supply to the LED block **92** is stopped, the power supply switch **94** is controlled to be turned on again, so that the light emission is continued in the normal-luminance mode in which the overpower (overcurrent) does not occur. Therefore, it is possible to realize the power redundancy.
[0123] Moreover, even in a configuration in which the plurality of display units **51** is connected like a daisy chain and video signals are sequentially transmitted, it is possible to reduce a risk that the video signal is not transmitted to the subsequent display unit **51** as the display unit **51** goes down due to overcurrent.
[0124] Note that, in the above description, the display unit **51** has been described as having a configuration including the LED drivers **121-1** to **121**-N and the LED arrays **122-1** to **122**-N, but the present embodiment is not limited thereto. The display unit **51** may include only one LED array **122-1** as the LED array, or may include only one LED driver **121-1** as the LED driver. Furthermore, the display unit **51** may not be a direct-view LED display, and may be, for example, a liquid crystal display using an LED backlight. Furthermore, the LEDs of the LED arrays **122-1** to **122**-N may be organic LEDs (OLEDs).
<Display Processing>
[0125] Next, display processing performed by the display system **11** in FIGS. **4** to **6** will be described with reference to the flowchart in FIG. **7**.
[0126] In step S**11**, the signal processing unit **78** receives an input of the video signal containing content data and the like supplied from the video server **31** via any of the HDMI terminal **72**, the DP terminal **73**, or the DVI terminal **74**, and the signal input IF **77**.
[0127] In step S**12**, the signal processing unit **78** converts a video format of the received video signal.
[0128] In step S**13**, the signal processing unit **78** determines whether or not an abnormality is detected from any of the display units **51-1** to **51**-*n* and display in the normal-luminance mode has been notified via the signal distribution unit **80** and the signal IFs **81-1** to **81**-*n*.
[0129] In step S**13**, in a case where it is determined that no abnormality is detected from any of the display units **51-1** to **51**-*n* and display in the normal-luminance mode has not been notified, the processing proceeds to step S**14**.
[0130] In step S**14**, the signal processing unit **78** receives the input of the control signal supplied from the MPU **76**, the control signal being supplied in accordance with a detail of an operation made on the PC **30**, and executes the signal processing for color temperature, contrast, brightness, and the like in the high-luminance mode.
[0131] In step S**17**, the signal processing unit **78** allocates and distributes the video signal subjected to the signal processing to the display units **51-1** to **51**-*n* of the video wall **33**.
[0132] In step S**18**, the signal processing unit **78** transmits and outputs the distributed video signal to each of the display units **51-1** to **51**-*n* of the corresponding video wall **33**.
[0133] In step S**19**, it is determined whether or not termination of the processing has been instructed, and in a case where the termination of the processing has not been instructed, the processing returns to step S**11**, and the subsequent processing is repeated.
[0134] Then, in step S**19**, in a case where it is determined that the termination has been instructed, the processing ends.
[0135] On the other hand, in step S**13**, in a case where it is determined that an abnormality is detected from any of the display units **51-1** to **51**-*n* and display in the normal-luminance mode has been notified, the processing proceeds to step S**15**.
[0136] In step S**15**, the signal processing unit **78** notifies the PC **30** of information indicating that

an abnormality has occurred in any of the display units **51-1** to **51**-*n* through the MPU **76**, the network IF **75**, and the LAN **71**. Thereby, the PC **30** presents the information indicating that an abnormality has occurred in any of the display units **51-1** to **51**-*n* to the user.
[0137] In step S**16**, the signal processing unit **78** receives the input of the control signal supplied from the MPU **76**, the control signal being supplied in accordance with a detail of an operation made on the PC **30**, and executes the signal processing for color temperature, contrast, brightness, and the like in the normal-luminance mode, and the processing proceeds to step S**17**.
[0138] In a case where no abnormality is detected in any of the display units **51-1** to **51**-*n* through the above series of processing, the video signal read from the video server **31** is subjected to the signal processing in the high-luminance mode, and is distributed and transmitted among the display units **51-1** to **51**-*n* constituting the video wall **33** so as to allow the display units **51-1** to **51**-*n* to display respective videos, thereby allowing the video wall **33** to display the entire video of the video content in the high-luminance mode.
[0139] Furthermore, when detection of an abnormality from any of the display units **51-1** to **51**-*n* is notified, the information indicating that an abnormality is detected is presented to the user of the PC **30**, so that the user can recognize that an abnormality has occurred in the display unit **51**.
[0140] Furthermore, when detection of an abnormality is notified from any of the display units **51-1** to **51**-*n*, the video signal read from the video server **31** is subjected to the signal processing in the normal-luminance mode, and is distributed and transmitted among the display units **51-1** to **51**-*n* constituting the video wall **33** so as to allow the display units **51-1** to **51**-*n* to display respective videos, thereby allowing the video wall **33** to display the entire video of the video content in the normal-luminance mode.
[0141] That is, in this case, as will be described below, the display unit **51** in which an abnormality is detected switches the display from the high-luminance mode to the normal-luminance mode and performs display. When the display unit **51** having no abnormality continues the display in the high-luminance mode, the display of only the display unit **51** in which an abnormality has occurred is performed in the normal-luminance mode, so that only the display in the display unit **51** in which an abnormality has occurred is darkly displayed.
[0142] However, in the present disclosure, when an abnormality is detected in any of the display units **51**, the display is switched from the high-luminance mode to the normal-luminance mode even in the display unit **51** without an abnormality, and all the display units **51** perform the display in the normal-luminance mode. Therefore, it is possible to continue the display while preventing only the display of the display unit **51** in which an abnormality has occurred from being darkly displayed.
[0143] Note that when an abnormality is detected in any of the display units **51**, the light emission mode in the display unit **51** having no abnormality remains in the high-luminance mode in driver control processing to be described below with reference to FIG. **8**.
[0144] However, since the processing for the video signal in the video wall controller **32** is set to the normal-luminance mode, all the video signals distributed and supplied to all the display units **51** of the video wall **33** are in the luminance range defined in the normal-luminance mode.
[0145] As a result, the light emission mode in the display unit **51** having no abnormality remains the high-luminance mode, but the video signal is generated in the luminance range defined in the normal-luminance mode. Therefore, the displayed image is also the image in the luminance range of the normal-luminance mode, and all the display units **51** display the images in the normal-luminance mode.
[0146] Note that a plurality of the video wall controllers **32** may be used to control the video wall **33**. At this time, the light emission mode may be controlled to be common among the plurality of video wall controllers **32** by communication between the video wall controllers **32** or communication via the PC **30**. As a result, even in a case where the video wall **33** is controlled using the plurality of video wall controllers **32**, all the display units **51** of the video wall **33** are

displayed in the normal-luminance mode, so that it is possible to continue the display while preventing only the display of the display unit **51** in which an abnormality has occurred from being darkly displayed.

<Driver Control Processing in Display Unit>

[0147] Next, the driver control processing performed in the display unit **51** will be described with reference to a flowchart in FIG. **8**.

[0148] Note that this processing is based on the assumption that the normal-luminance power supplies **93-1** and **93-2** are subjected to sharing control, and the driver power supply output is supplied from the terminals **93*d*-1** and **93*d*-2** to the LED block **92** via the power supply switch **94**, and the system power supply output is supplied from the terminals **93*s*-1** and **93*s*-2** to the driver control circuit **91** on the basis of the AC power supplied from the AC input **90**.

[0149] In step S**31**, the signal processing unit **112** in the driver control circuit **91** of the display unit **51** sets the light emission mode to the high-luminance mode.

[0150] In step S**32**, the monitoring unit **110** in the driver control circuit **91** of the display unit **51** controls the power supply switch **94** to be turned on.

[0151] In step S**33**, the monitoring unit **110** determines whether or not the abnormality detection signal indicating that an abnormality has occurred is supplied from at least one of the terminals **93*e*-1** and **93*e*-2** of the normal-luminance power supplies **93-1** and **93-2**.

[0152] In step S**33**, in a case where it is determined that the abnormality detection signal is not supplied from any of the terminals **93*e*-1** and **93*e*-2** of the normal-luminance power supplies **93-1** and **93-2**, the processing proceeds to step S**34**.

[0153] In step S**34**, the signal processing unit **112** receives the input of the video signal distributed by and supplied from the video wall controller **32** in units of rows via the signal IF **111**.

[0154] In step S**35**, the signal processing unit **112** executes the video signal processing of correcting color, luminance, and the like corresponding to each of the display units **51** for the video signal in units of rows distributed among the display units **51** in the set light emission mode. That is, when the light emission mode is the high-luminance mode, the video signal processing is performed in the high-luminance mode, and when the light emission mode is the normal-luminance mode, the video signal processing is performed in the normal-luminance mode.

[0155] In step S**36**, the signal processing unit **112** allocates the video signal in units of rows subjected to the video signal processing to the LED drivers **121-1** to **121**-N of the LED block **92**, and transmits the video signals via the corresponding output IFs **114-1** to **114**-N.

[0156] In step S**37**, the LED drivers **121-1** to **121**-N of the LED block **92** execute LED drive control processing on the basis of the video signal in units of rows, and each of the LED arrays **122-1** to **122**-N displays a video in units of rows with appropriate luminance.

[0157] In step S**38**, it is determined whether or not termination of the processing has been instructed, and in a case where the termination of the processing has not been instructed, the processing returns to step S**33**, and the subsequent processing is repeated.

[0158] Then, in step S**38**, when the termination of the processing is instructed, the processing ends.

[0159] That is, in the case where no abnormality is detected in any of the normal-luminance power supplies **93-1** and **93-2**, the LED arrays **122-1** to **122**-N display the video in units of rows with appropriate luminance in the high-luminance mode.

[0160] On the other hand, in step S**33**, in the case where it is determined that an abnormality is detected in any of the normal-luminance power supplies **93-1** and **93-2**, the processing proceeds to step S**39**.

[0161] In step S**39**, the power supply switch **94** turns off the connection on the basis of the abnormality detection signal, stops the power supply to the LED block **92**, and stops the display.

[0162] In step S**40**, the monitoring unit **110** determines whether or not an abnormality is detected in all the normal-luminance power supplies **93-1** and **93-2**.

[0163] In step S**40**, in a case where an abnormality is not detected in all the normal-luminance power supplies **93-1** and **93-2** and an abnormality is detected in either one of them, the processing proceeds to step S**41**.

[0164] In step S**41**, the monitoring unit **110** notifies the signal processing unit **112** of the information indicating that an abnormality is detected. With this notification, the signal processing unit **112** sets the light emission mode to the normal-luminance mode.

[0165] In step S**42**, the monitoring unit **110** supplies the re-energization signal to the power supply switch **94**, controls the power supply switch **94** to be turned on, resumes the power supply to the LED block **92**, and resumes the display.

[0166] In step S**43**, the signal processing unit **112** controls the signal IF **111** to notify the video wall controller **32** that there is an abnormality in some normal-luminance power supply **93** of the display unit **51** and the LED block **92** is displayed in the normal-luminance mode, and the processing returns to step S**38**.

[0167] That is, when an abnormality is detected in any of the normal-luminance power supplies **93-1** and **93-2**, the LED arrays **122-1** to **122**-N display the video in units of rows with appropriate luminance in the normal-luminance mode in the subsequent processing.

[0168] Furthermore, at this time, the video wall controller **32** is notified that an abnormality is detected in any of the normal-luminance power supplies **93-1** and **93-2** and the display is performed in the normal-luminance mode, and moreover, the PC **30** is notified of these pieces of information through the video wall controller **32**, and the pieces of information are presented in the PC **30**.

[0169] Moreover, in step S**40**, in the case where an abnormality is detected in all the normal-luminance power supplies **93-1** and **93-2**, the processing proceeds to step S**44**.

[0170] In step S**44**, the signal processing unit **112** controls the signal IF **111** to notify the video wall controller **32** that there is an abnormality in all the normal-luminance power supplies **93** of the display unit **51**, the display by the LED block **92** is stopped, and the black screen state is set, and the processing ends.

[0171] That is, when an abnormality is detected in both of the normal-luminance power supplies **93-1** and **93-2**, the LED arrays **122-1** to **122**-N are in an undisplayable state in the subsequent processing, and thus the video wall controller **32** is notified of the black screen state.

[0172] Furthermore, at this time, the video wall controller **32** is notified that an abnormality is detected in both the normal-luminance power supplies **93-1** and **93-2**, the display is stopped, and the black screen state is set, and moreover, the PC **30** is notified of these pieces of information through the video wall controller **32**, and the pieces of information are presented in the PC **30**.

[0173] By the above processing, in the case where an abnormality is detected in any of the normal-luminance power supplies **93-1** and **93-2**, the power supply to the LED arrays **122-1** to **122-**N is stopped and the display is stopped, and the power supply is resumed after the light emission mode is switched from the high-luminance mode to the normal-luminance mode.

[0174] Since a period from the stop to the restart of the power supply is a period corresponding to a display time of several frames required for switching the light emission mode from the high-luminance mode to the normal-luminance mode, the display is substantially stopped only for an extremely short period, and thus the display is switched to the normal-luminance mode and continued.

[0175] Furthermore, as described above, when the display is performed in the normal-luminance mode, information of the display is notified to the video wall controller **32**, and the video signal to be generated in the video wall controller **32** is generated in the normal-luminance mode in response to the notification, so that the video signal is displayed in the normal-luminance mode in all the display units **51** including the display unit **51** in which no abnormality has occurred.

[0176] Thereby, it is possible to avoid a display state in which only the display unit **51** in which an abnormality has occurred is displayed in the normal-luminance mode, and thus looks darker than the display unit **51** displayed in another high-luminance mode.

[0177] Moreover, when an abnormality is detected in any of the display units **51**, the information indicating that the abnormality is detected is supplied from the video wall controller **32** to the PC **30** and is presented to the user, so that the user can recognize that an abnormality is detected in the display unit **51** even if the display is continued in the normal-luminance mode.
[0178] Note that, in the above configuration, an example has been described in which, when the abnormality detection signal is supplied, the power supply switch **94** controls itself to be turned off, changes the high-luminance mode to the normal-luminance mode, and then controls itself to be turned on on the basis of the re-energization signal supplied from the monitoring unit **110**.
[0179] However, the abnormality detection signal may not be supplied to the power supply switch **94**, and on or off of the power supply switch **94** may be controlled by the monitoring unit **110**.
[0180] That is, the abnormality detection signal may be supplied only to the monitoring unit **110** of the driver control circuit **91**, and the monitoring unit **110** may turn off the power supply switch **94** when detecting the abnormality detection signal, and may turn on the power supply switch **94** again after the light emission mode is switched from the high-luminance mode to the normal-luminance mode. In this way, the control system of the power supply switch **94** can be integrated by the monitoring unit **110**.
3. Application Example
[0181] In the above description, in the case where the abnormality detection signal is detected in any of the normal-luminance power supplies **93-1** and **93-2**, the power supply switch **94** is turned off to stop the power supply, the display mode is switched from the high-luminance mode to the normal-luminance mode, and then the power supply switch **94** is turned on to resume the power supply.
[0182] However, in the case where the display mode is the high-luminance mode, even if an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, the overpower (overcurrent) does not occur even if the display is continued when the video signal itself supplied from the video wall controller **32** is in the luminance range displayable in the normal-luminance mode.
[0183] Therefore, even if an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, it is not always necessary to turn off the power supply switch **94** and switch the display mode from the high-luminance mode to the normal-luminance mode when the video signal itself is in the luminance range displayable in the normal-luminance mode.
[0184] Therefore, in the case where the light emission mode is the high-luminance mode, the abnormality detection signal may not be supplied to the power supply switch **94** even if the abnormality detection signal is detected in any of the normal-luminance power supplies **93-1** and **93-2** when the video signal is in the luminance range displayable even in the normal-luminance mode, and the abnormality detection signal may be supplied to the power supply switch **94** when the video signal is in the luminance range undisplayable in the normal-luminance mode.
[0185] FIG. **9** illustrates a configuration example of the display unit **51** configured not to supply the abnormality detection signal to the power supply switch **94** even if the abnormality detection signal is detected in any of the normal-luminance power supplies **93-1** and **93-2** when the video signal is in the luminance range displayable even in the normal-luminance mode, and to supply the abnormality detection signal to the power supply switch **94** when the video signal is in the luminance range undisplayable in the normal-luminance mode, in the case of the high-luminance mode.
[0186] In the display unit **51** of FIG. **9**, components having the same functions as those of the display unit **51** of FIG. **5** are denoted by the same reference numerals, and description thereof will be appropriately omitted.
[0187] The display unit **51** of FIG. **9** is different from the display unit **51** of FIG. **5** in that a monitoring unit **110′** is provided instead of the monitoring unit **110**, and an abnormality detection signal switch **151** is newly provided.

[0188] The abnormality detection signal switch **151** is an opening/closing switch including a MOS-FET or the like whose on/off is controlled by the monitoring unit **110**′, and is a switch that switches on/off of connection between the terminals **93**_e_**-1** and **93**_e_**-2** of the normal-luminance power supplies **93-1** and **93-2** and the power supply switch **94**. Note that the abnormality detection signal switch **151** may have a configuration other than the opening/closing switch as long as the on/off is controlled by the monitoring unit **110**′, and may be configured by, for example, a logic circuit.
[0189] The monitoring unit **110**′ has the same basic function as the monitoring unit **110**, but further determines whether or not the video signals supplied from the signal processing unit **112** to the LED block **92** via the output IFs **114-1** to **114**-N are in the luminance range displayable even in the normal-luminance mode, and controls the abnormality detection signal switch **151** to be turned off when it is determined that the video signals are in the displayable luminance range and controls the abnormality detection signal switch **151** to be turned on when it is determined that the video signal is in the undisplayable luminance range.
[0190] According to such control, in the high-luminance mode, even if an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, when the video signal itself is in the luminance range displayable in the normal-luminance mode, the abnormality detection signal is not supplied to the power supply switch **94**. Therefore, the power supply is not stopped, and the display mode remains in the high-luminance mode.
[0191] Furthermore, since the video wall controller **32** is not notified of the abnormality, the signal processing unit **78** does not need to perform the signal processing by setting the video signals to be supplied to all the display units **51** to the normal-luminance mode.
<Abnormality Detection Signal Switch Control Processing>
[0192] Next, abnormality detection signal switch control processing by the display unit **51** in FIG. **9** will be described with reference to the flowchart in FIG. **10**.
[0193] In step S**71**, the monitoring unit **110**′ controls the abnormality detection signal switch **151** to be turned on.
[0194] In step S**72**, the monitoring unit **110**′ acquires the video signals in units of rows subjected to the video signal processing, which are to be output from the signal processing unit **112** to the output IFs **114-1** to **114**-N.
[0195] In step S**73**, the monitoring unit **110**′ determines whether or not the acquired video signals subjected to the video signal processing in units of rows are in the luminance range displayable in the normal-luminance mode.
[0196] In step S**73**, in the case where it is determined that the video signals are in the luminance range displayable in the normal-luminance mode, the processing proceeds to step S**74**.
[0197] In step S**74**, the monitoring unit **110**′ sets the abnormality detection signal switch **151** to off.
[0198] On the other hand, in step S**73**, in the case where it is determined that the video signals are not in the luminance range displayable in the normal-luminance mode, the processing proceeds to step S**75**.
[0199] In step S**75**, the monitoring unit **110**′ sets the abnormality detection signal switch **151** to on.
[0200] In step S**76**, it is determined whether or not termination of the processing has been instructed, and the processing returns to step S**72** in a case where termination is not instructed. That is, the processing of steps S**72** to S**76** is repeated until termination is instructed.
[0201] Then, in step S**76**, when it is determined that the instruction on the end is given, the processing ends.
[0202] According to the above processing, in the high-luminance mode, even if an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, when the video signal itself is in the luminance range displayable in the normal-luminance mode, the abnormality detection signal is not supplied to the power supply switch **94**. Therefore, the power supply is not stopped, and the display mode can be kept in the high-luminance mode.
[0203] Furthermore, since the video wall controller **32** is not notified of the abnormality, the signal

processing unit **78** does not need to perform the signal processing by setting the video signals to be supplied to all the display units **51** to the normal-luminance mode.

[0204] As a result, even if an abnormality occurs in any of the normal-luminance power supplies **93-1** and **93-2**, it is possible to realize power supply redundancy, and even if an abnormality occurs in the power supply in any of the display units **51**, it is possible to reduce the influence on the video wall controller **32** and the other display units **51**.

[0205] Note that, although an example in which two normal-luminance power supplies **93** are provided in one display unit **51** has been described, the number of normal-luminance power supplies may be more than two. In that case, in a case where there are two or more normal-luminance power supplies **93** having no abnormality, the signal processing unit **112** can be used in the high-luminance mode, and switches the light emission mode to the normal-luminance mode when the number of normal-luminance power supplies **93** having no abnormality becomes one to enable generation of the video signal.

[0206] Note that the present disclosure may also have the following configurations. [0207] <1> A display system including: [0208] a driver configured to drive a display element on the basis of a video signal; [0209] a plurality of power supplies configured to supply power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range; and [0210] a driver control unit configured to control the driver so as to switch a luminance mode of the display element, in which [0211] the driver control unit controls the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by power supply from the plurality of power supplies, and [0212] performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of detection of an abnormality in any of the plurality of power supplies. [0213] <2> The display system according to <1>, further including: [0214] a power supply switch configured to switch on or off of the power supply from the plurality of power supplies to the driver, in which [0215] the driver control unit performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode after the power supply switch is turned off on the basis of detection of an abnormality in any of the plurality of power supplies. [0216] <3> The display system according to <2>, in which [0217] the driver control unit controls the power supply switch to be turned on, and [0218] on the basis of the detection of an abnormality in any of the plurality of power supplies, [0219] the power supply switch turns off the power supply from the plurality of power supplies to the driver, and [0220] the driver control unit switches from the second luminance mode to the first luminance mode and performs control so that the display element is driven by the driver, and then controls the power supply switch to be turned on to resume the power supply from the plurality of power supplies to the driver. [0221] <4> The display system according to <3>, in which [0222] the driver control unit controls the driver to drive the display element in the second luminance mode on the basis of a fact that there is no abnormality in any of the plurality of power supplies. [0223] <5> The display system according to any of <1> to <4>, in which, [0224] in a case where the power supply is a single unit, [0225] the power supply of the power required for driving the display element is sometimes disabled in the second luminance mode, and [0226] the power supply of the power required for driving the display element is always suppliable in the first luminance mode. [0227] <6> The display system according to any of <1> to <5>, in which [0228] the plurality of power supplies is subjected to sharing control so that current values become the same, and supplies the power to the driver in parallel. [0229] <7> The display system according to any of <1> to <6>, in which [0230] the plurality of power supplies includes two of the power supplies. [0231] <8> The display system according to <3>, in which [0232] the power supply further includes an abnormality detection unit that detects an own abnormality, [0233] the abnormality detection unit outputs an abnormality detection signal on the basis of the detection of the own abnormality, and [0234] on the basis of the output of the abnormality detection signal, [0235] the power supply switch turns off the power supply from the plurality of power supplies to the driver, and [0236] the driver control unit controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and then controls the power supply switch to be turned on again. [0237] <9> The display system according to <8>, in which [0238] the driver control unit controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and then transmits a re-energization signal to the power supply switch, and [0239] the power supply switch is controlled to be turned on again on the basis of the re-energization signal. [0240] <10> The display system according to <3>, in which [0241] the power supply further includes an abnormality detection unit that detects an own abnormality and outputs an abnormality detection signal on the basis of the detection of the own abnormality, and [0242] on the basis of the output of the abnormality detection signal, [0243] the driver control unit controls the power supply from the plurality of power supplies of the power supply switch to the driver to be turned off, then controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and controls the power supply switch to be turned on again. [0244] <11> The display system according to <3>, in which [0245] the power supply further includes: an abnormality detection unit configured to detect an own abnormality and output an abnormality detection signal on the basis of the detection of the own abnormality; and [0246] an abnormality detection signal switch configured to control supply of the abnormality detection signal to the power supply switch supplied from the abnormality detection unit, in which [0247] the driver control unit controls on or off of the abnormality detection signal switch on the basis of luminance of the video signal subjected to video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode. [0248] <12> The display system according to <11>, in which [0249] the driver control unit controls on or off of the abnormality detection signal switch on the basis of whether or not the luminance of the video signal subjected to the video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode, is within the first luminance range. [0250] <13> The display system according to <12>, in which [0251] the driver control unit controls the abnormality detection signal switch to be turned off on the basis of a fact that the luminance of the video signal subjected to the video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode, is within the first luminance range, and controls the abnormality detection signal switch to be turned on on the basis of a fact that the luminance of the video signal subjected to the video signal processing is not within the first luminance range. [0252] <14> The display system according to <3>, further including: [0253] a plurality of display units including the plurality of power supplies, the power supply switch, and the driver control unit; and [0254] a controller configured to perform the video signal processing for the video signal, and distributes and supplies the video signal to the plurality of display units, in which [0255] the driver control unit notifies the controller that the abnormality is detected in any of the plurality of power supplies on the basis of the detection of the abnormality in any of the plurality of power supplies. [0256] <15> The display system according to <14>, in which [0257] the controller performs the video signal processing for the video signal in the first luminance mode and distributes and supplies the video signal to the plurality of display units on the basis of the notification that the abnormality is detected in any of the plurality of power supplies from the driver control unit of any of the plurality of display units. [0258] <16> The display system according to <14>, in which [0259] the controller outputs information for presenting that the abnormality is detected in any of the plurality of power supplies on the basis of the notification that the abnormality is detected in any of the plurality of power supplies from the driver control unit of any of the plurality of display units. [0260] <17> The display system according to <14>, in which [0261] the driver control unit notifies the controller that the abnormality is detected in all of the plurality of power supplies and display is disabled on the basis of the detection of the abnormality in all of the plurality of power supplies. [0262] <18> The display system according to <14>, in which [0263] the plurality of display units is arranged in an array and constitute a video wall, and [0264] an image formed by the video signal is displayed as one image on the entire video wall. [0265] <19> The display system according to <18>, in which [0266] a first display unit included in the plurality of display units is connected to a second display unit included in the plurality of display units, and [0267] the controller supplies the video signal to the second display unit via the first display unit. [0268] <20> A method of operating a display system including [0269] a driver that drives a display element on the basis of a video signal, [0270] a plurality of power supplies that supplies power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range, and [0271] a driver control unit that controls the driver so as to switch a luminance mode of the display element, the method including: [0272] by the driver control unit, [0273] controlling the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by the power supply from the plurality of power supplies; and [0274] performing control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of detection of an abnormality in any of the plurality of power supplies, in which [0275] the driver control unit drives the display element in the second luminance mode by the power supply from the plurality of power sources, and [0276] performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on the basis of the detection of the abnormality in any of the plurality of power supplies.

REFERENCE SIGNS LIST

[0277] **11** Display system [0278] **30** PC [0279] **31** Video server [0280] **32** Video wall controller [0281] **33** Video wall [0282] **51** and **51-1** to **51-***n* Display unit [0283] **78** Signal processing unit [0284] **90** AC input [0285] **91** Driver control circuit [0286] **92** LED block [0287] **93**, **93-1**, **93-2** Normal-luminance power supply [0288] **93***e***-1**, **93***e***-2**, **93***s***-1**, **93***s***-2**, **93***d***-1**, **93***d***-2** Terminal [0289] **94** Power supply switch [0290] **110**, **110**′ Monitoring unit [0291] **112** Signal processing unit [0292] **121** and **121-1** to **121-**N Drive circuit [0293] **122** Pixel array [0294] **131**, **131-1**, **131-2** Abnormality detection unit [0295] **132**, **132-1**, **132-2** Driver power supply output [0296] **133**, **133-1**, **133-2** System power supply output [0297] **151** Abnormality detection signal switch

Claims

1. A display system comprising: a driver configured to drive a display element on a basis of a video signal; a plurality of power supplies configured to supply power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range; and a driver control unit configured to control the driver so as to switch a luminance mode of the display element, wherein the driver control unit controls the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by power supply from the plurality of power supplies, and performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on a basis of detection of an abnormality in any of the plurality of power supplies.

2. The display system according to claim 1, further comprising: a power supply switch configured to switch on or off of the power supply from the plurality of power supplies to the driver, wherein the driver control unit performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode after the power supply switch is turned off on a basis of detection of an abnormality in any of the plurality

of power supplies.
3. The display system according to claim 2, wherein the driver control unit controls the power supply switch to be turned on, and on a basis of the detection of an abnormality in any of the plurality of power supplies, the power supply switch turns off the power supply from the plurality of power supplies to the driver, and the driver control unit switches from the second luminance mode to the first luminance mode and performs control so that the display element is driven by the driver, and then controls the power supply switch to be turned on to resume the power supply from the plurality of power supplies to the driver.
4. The display system according to claim 3, wherein the driver control unit controls the driver to drive the display element in the second luminance mode on a basis of a fact that there is no abnormality in any of the plurality of power supplies.
5. The display system according to claim 1, wherein, in a case where the power supply is a single unit, the power supply of the power required for driving the display element is sometimes disabled in the second luminance mode, and the power supply of the power required for driving the display element is always suppliable in the first luminance mode.
6. The display system according to claim 1, wherein the plurality of power supplies is subjected to sharing control so that current values become the same, and supplies the power to the driver in parallel.
7. The display system according to claim 1, wherein the plurality of power supplies includes two of the power supplies.
8. The display system according to claim 3, wherein the power supply further includes an abnormality detection unit that detects an own abnormality, the abnormality detection unit outputs an abnormality detection signal on a basis of the detection of the own abnormality, and on a basis of the output of the abnormality detection signal, the power supply switch turns off the power supply from the plurality of power supplies to the driver, and the driver control unit controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and then controls the power supply switch to be turned on again.
9. The display system according to claim 8, wherein the driver control unit controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and then transmits a re-energization signal to the power supply switch, and the power supply switch is controlled to be turned on again on a basis of the re-energization signal.
10. The display system according to claim 3, wherein the power supply further includes an abnormality detection unit that detects an own abnormality and outputs an abnormality detection signal on a basis of the detection of the own abnormality, and on a basis of the output of the abnormality detection signal, the driver control unit controls the power supply from the plurality of power supplies of the power supply switch to the driver to be turned off, then controls the driver to switch the display element from the second luminance mode to the first luminance mode to drive the display element, and controls the power supply switch to be turned on again.
11. The display system according to claim 3, wherein the power supply further includes: an abnormality detection unit configured to detect an own abnormality and output an abnormality detection signal on a basis of the detection of the own abnormality; and an abnormality detection signal switch configured to control supply of the abnormality detection signal to the power supply switch supplied from the abnormality detection unit, and the driver control unit controls on or off of the abnormality detection signal switch on a basis of luminance of the video signal subjected to video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode.
12. The display system according to claim 11, wherein the driver control unit controls on or off of the abnormality detection signal switch on a basis of whether or not the luminance of the video signal subjected to the video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode, is within the first

luminance range.

13. The display system according to claim 12, wherein the driver control unit controls the abnormality detection signal switch to be turned off on a basis of a fact that the luminance of the video signal subjected to the video signal processing, the video signal being generated by executing the video signal processing for the video signal in the second luminance mode, is within the first luminance range, and controls the abnormality detection signal switch to be turned on on a basis of a fact that the luminance of the video signal subjected to the video signal processing is not within the first luminance range.

14. The display system according to claim 3, further comprising: a plurality of display units including the plurality of power supplies, the power supply switch, and the driver control unit; and a controller configured to perform the video signal processing for the video signal, and distributes and supplies the video signal to the plurality of display units, wherein the driver control unit notifies the controller that the abnormality is detected in any of the plurality of power supplies on a basis of the detection of the abnormality in any of the plurality of power supplies.

15. The display system according to claim 14, wherein the controller performs the video signal processing for the video signal in the first luminance mode and distributes and supplies the video signal to the plurality of display units on a basis of the notification that the abnormality is detected in any of the plurality of power supplies from the driver control unit of any of the plurality of display units.

16. The display system according to claim 14, wherein the controller outputs information for presenting that the abnormality is detected in any of the plurality of power supplies on a basis of the notification that the abnormality is detected in any of the plurality of power supplies from the driver control unit of any of the plurality of display units.

17. The display system according to claim 14, wherein the driver control unit notifies the controller that the abnormality is detected in all of the plurality of power supplies and display is disabled on a basis of the detection of the abnormality in all of the plurality of power supplies.

18. The display system according to claim 14, wherein the plurality of display units is arranged in an array and constitute a video wall, and an image formed by the video signal is displayed as one image on the entire video wall.

19. The display system according to claim 18, wherein a first display unit included in the plurality of display units is connected to a second display unit included in the plurality of display units, and the controller supplies the video signal to the second display unit via the first display unit.

20. A method of operating a display system including a driver that drives a display element on a basis of a video signal, a plurality of power supplies that supplies power capable of driving the display element to the driver in a first luminance mode in which the display element is driven in a first luminance range, and a driver control unit that controls the driver so as to switch a luminance mode of the display element, the method comprising: by the driver control unit, controlling the driver to drive the display element in a second luminance mode in which the display element is driven in a second luminance range higher in luminance than the first luminance range by the power supply from the plurality of power supplies; and performing control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on a basis of detection of an abnormality in any of the plurality of power supplies, wherein the driver control unit drives the display element in the second luminance mode by the power supply from the plurality of power sources, and performs control to switch from the second luminance mode to the first luminance mode so that the driver drives the display element in the first luminance mode on a basis of the detection of the abnormality in any of the plurality of power supplies.